Urban Heat Islands

Student sheet

Name: _____ **Date:** _____

Safety: Do not sit on hot asphalt, and keep away from cars.

1. What I see on the screen

The screen shows **Temperature**, **Humidity**, the **Capture** button, the chart and, at the analysis, a table with a **Corrected** column. **This sheet is the record.**

I measure the air above the surface, not the surface. Asphalt can be 60 °C, but the air above it, much less.

The protocol — five rules, at every location: (1) a height of 1.2 m, the same everywhere, measured with the tape; **(2)** a white screen above the sensor, *including* in the shade (otherwise the box measures its own casing); **(3) a full two minutes**, timed; **(4)** everyone at **two metres** while it settles; **(5) the time noted** with the number.

2. What I do on my own

Weather: _____ **Start:** _____ **End:** _____

First I predict, on the sheet and in the app. I do not change it. 1. The warmest location will be _____, the coolest _____. **2.** The difference between them will be _____ °C. **3.** Indoors will be ☐ warmer ☐ cooler than outdoors, because _____

Now I measure. Row 7 (the return to spot 1) **is compulsory.**

#	Location (surface, sun or shade)	Time	T (°C)	RH (%)	Corrected
1	Grass in shade — **reference**				
2	Asphalt in sun				
3	Indoors (classroom or corridor)				
4					
5					
6					
7	Grass in shade — **return**				

As an observation only, not in the table: above the asphalt, at 10 cm, $T =$ _____ °C, versus _____ °C at the protocol height.

The map. Write the *corrected* temperature in each location's box, then colour blue for the coolest and red for the warmest. The schoolyard plan, if you received one, is glued to the back of the sheet.

THE BUILDING (indoors) no. ___ $T =$_____°C	**ASPHALT IN SUN** no. ___ $T =$_____°C	**ASPHALT IN SHADE** no. ___ $T =$_____°C
GRASS IN SHADE (ref.) no. ___ $T =$_____°C	**GRASS IN SUN** no. ___ $T =$_____°C	**BESIDE THE SUNLIT WALL** no. ___ $T =$_____°C

Calibration, before setting off: the two boxes side by side, in the shade, 5 min. My box _____ °C, the reference _____ °C, **the offset** _____ °C. (With a single box: I compare it with a thermometer.)

The drift (how much the whole location warmed up while we walked) = row 7 − row 1 = _____ − _____ = _____ °C. **The correction** for a stop made t minutes after row 1, on a walk of T minutes, is the drift $\times\, t/T$; example: $0.8 \times 15/30 = 0.40$, so 26.1 becomes 25.7 °C (as in the **Corrected** column).

ΔT (grass in shade → asphalt in sun) = _____ − _____ = _____ °C Warmest: _____

ΔT (indoors → outdoors) = _____ − _____ = _____ °C Coolest: _____

3. The conclusion I write

My drift: _____ °C. **The offset:** _____ °C. **Biggest difference between locations:** _____ °C. So the difference ☐ is detectable ☐ **is not detectable** with our setup (not detectable if the drift or the offset is bigger than it). *"Not detectable" does not mean "does not exist".*

I predicted that, and the data show that _____

_____

The coolest location was the coolest because _____

_____

I propose that our schoolyard should _____

_____

Exit ticket

One thing that surprised you today, and one thing you would do differently if you repeated the measurement.

_____